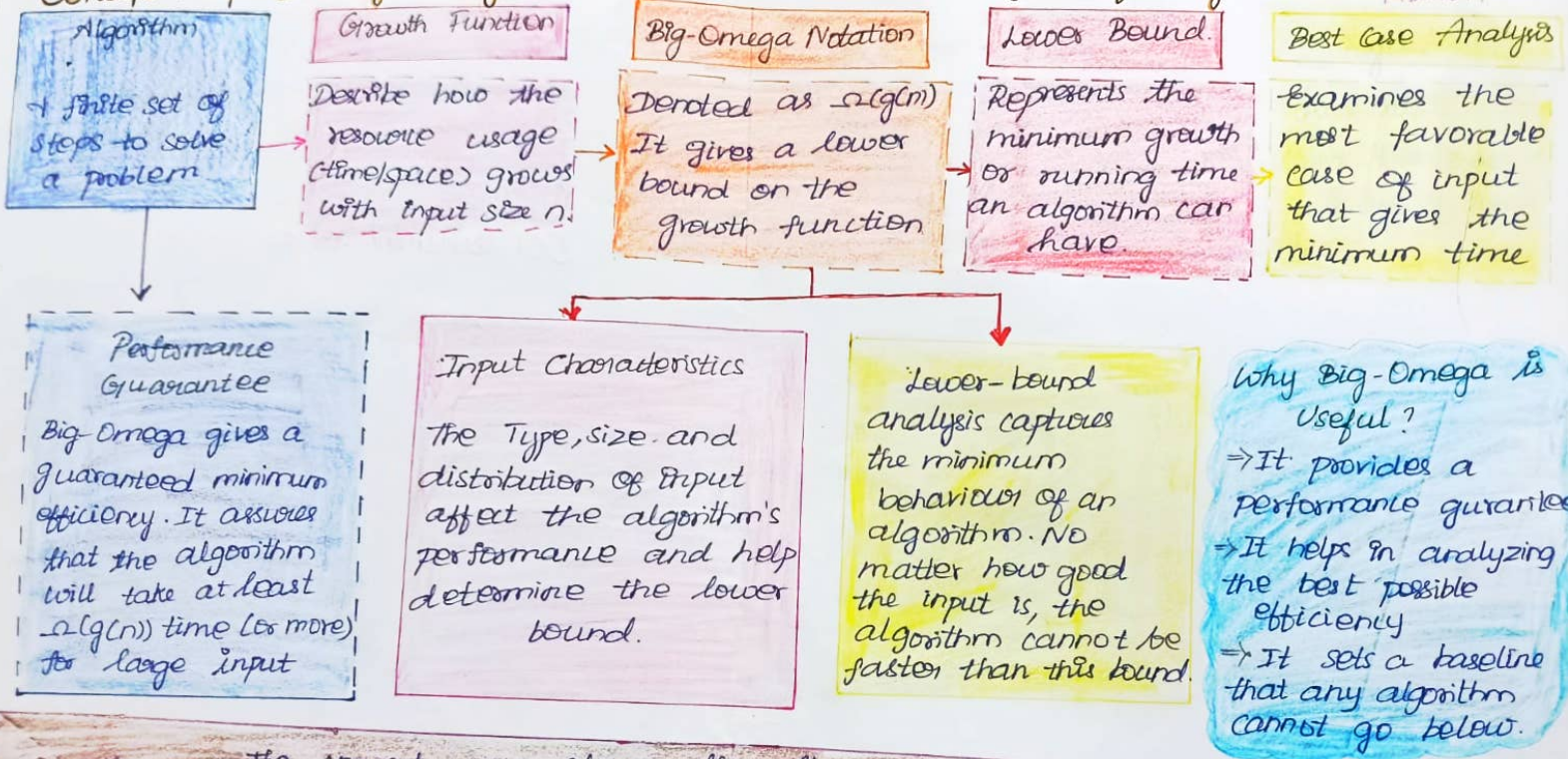
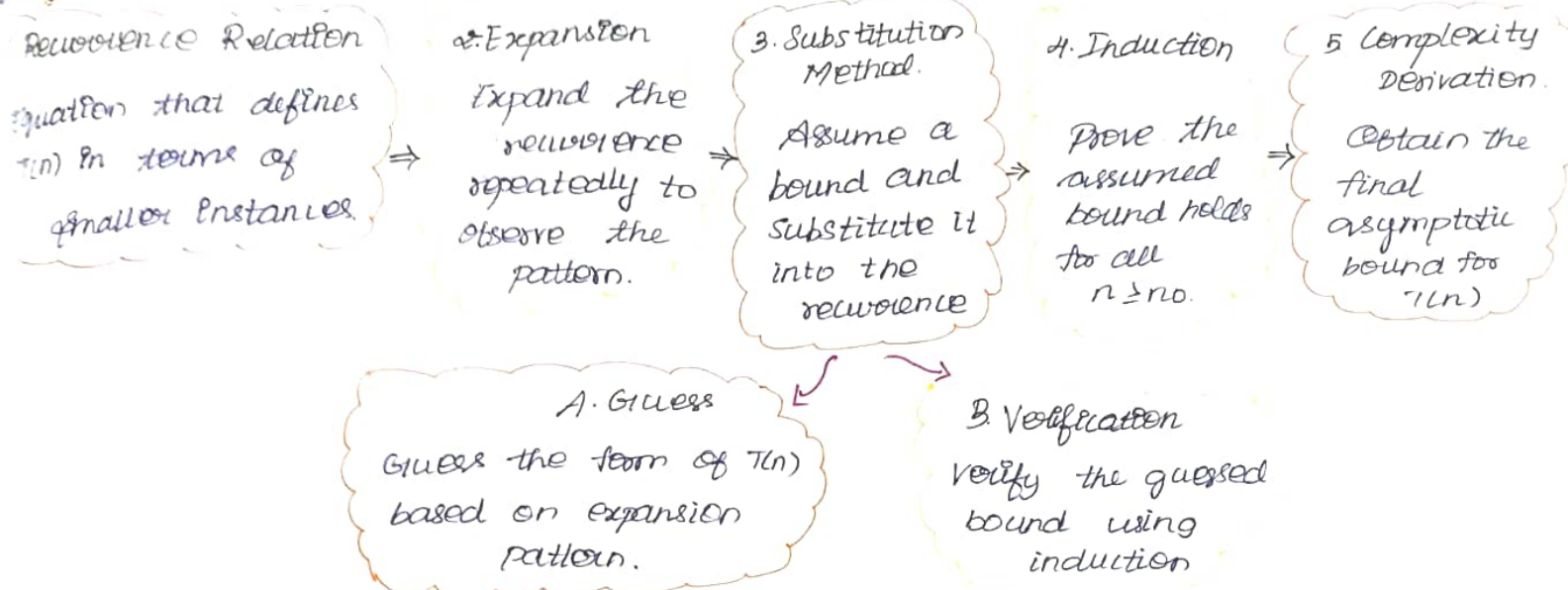


Concept Map On Big-Omega Notation and Lower-Bound Analysis of Algorithms. P. Krishnapriya 192511279



Overall Interpretation
The concept map shows the flow from an algorithm to its guaranteed minimum performance. It connects the mathematical notion (Big-Omega) with practical analysis (best-case & input effects) to provide a balanced understanding of algorithm behavior from the lower-bound perspective.

Solving Recurrence Relations Using the Substitution Method and Mathematical Induction



Why induction is important?

- * It confirms the guessed bound is true for all $n \geq n_0$.
- * It removes doubt and make the result mathematically correct.

How the map **help**?

It gives a step by step systematic approach to solve recurrence and derive time complexity.

Step by Step Flow:

- * Start with the recurrence
- * Expand to find the pattern
- * Guess the bound
- * Use substitution to test the guess
- * Prove it using induction
- * Derive the final complexity